AI Ka Chilla - Assignment Summary

This document provides a comprehensive overview of the assignments completed during the AI Ka Chilla course. Each section corresponds to a specific assignment, detailing the tasks accomplished, tools used, and outcomes achieved.

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1. **Python and Other Software Integration**

Python & GitBash:

```
(base)
UMS@DESKTOP-QAIDGDP MINGW64 ~
$ python --version
Python 3.13.9
(base)
UMS@DESKTOP-QAIDGDP MINGW64 ~
$ conda --version
conda 25.9.1
(base)
UMS@DESKTOP-QAIDGDP MINGW64 ~
$ bash --version
GNU bash, version 5.2.37(1)-release (x86_64-pc-msys)
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License GPLv3+: GNU GPL version 3 or later <http://gnu.org/licenses/gpl.html>

This is free software; you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.
(base)
UMS@DESKTOP-QAIDGDP MINGW64 ~
$ |
```

Anaconda integration:

```
Anaconda Prompt - "E:\Anaconda\condabin\conda.bat"  activate data_env

(base) C:\Users\UMS>conda --version
conda 25.9.1

(base) C:\Users\UMS>conda env list

# conda environments:
#
# *  -> active
# + -> frozen
base                  *   E:\Anaconda
ai_agents                 E:\Anaconda\envs\ai_agents
ai_env                    E:\Anaconda\envs\ai_env
appenv                    E:\Anaconda\envs\appenv
arduino_env               E:\Anaconda\envs\arduino_env
automation_env            E:\Anaconda\envs\automation_env
data_env                  E:\Anaconda\envs\data_env
machine_deep_env          E:\Anaconda\envs\machine_deep_env
website_env               E:\Anaconda\envs\website_env
webtools_env              E:\Anaconda\envs\webtools_env
ai_agents                 e:\Anaconda\envs\ai_agents


(base) C:\Users\UMS>conda activate data_env

(data_env) C:\Users\UMS>_
```

Jupyter Notebook & python libraries:

01_jupiter.ipynb > print(sys.version)

Generate + Code + Markdown | Run All Restart Clear All Outputs ··· data_env (Python 3.10.19)

Generate + Code + Markdown

```python
import sys
import pandas as pd
import matplotlib as plt
import seaborn as sns
import numpy as np
import scipy as sci
import sklearn as sk
import plotly as py
```

[12] ✓ 0.0s Python

```python
print(sys.version)
print(pd.__version__)
print(plt.__version__)
print(sns.__version__)
print(np.__version__)
print(sci.__version__)
print(sk.__version__)
print(py.__version__)
```

[13] ✓ 0.0s Python

```
3.10.19 | packaged by Anaconda, Inc. | (main, Oct 21 2025, 16:41:31) [MSC v.1929 64 bit (AMD64)]
2.3.3
3.10.8
0.13.2
2.2.6
1.15.3
1.7.2
6.5.2
```

2. **Python & Markdown Practice**

Python
Markdown

3. **My Portfolio**

My_Portfolio

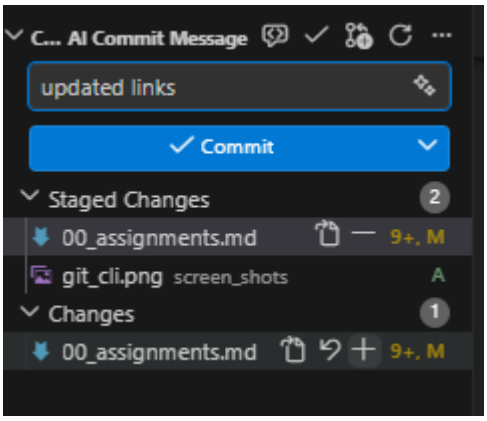
4. **Git & Github (CLI, VSCode)**

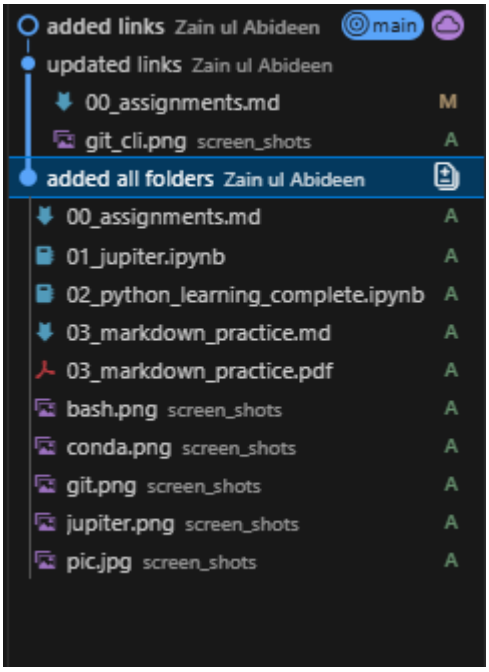
```
C:\Users\UMS>git --version
git version 2.52.0.windows.1

C:\Users\UMS>_
```

Local Repo to Github Repo in CLI:

```
(base)
UMS@DESKTOP-QAIDGDP MINGW64 /e/programming/AI_ka_chilla_assignments
$ git init
Initialized empty Git repository in E:/programming/AI_ka_chilla_assignments/.git/
(base)
UMS@DESKTOP-QAIDGDP MINGW64 /e/programming/AI_ka_chilla_assignments (master)
$ git add .
warning: in the working copy of '01_jupiter.ipynb', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of '02_python_learning_complete.ipynb', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of '03_markdown_practice.md', LF will be replaced by CRLF the next time Git touches it
(base)
UMS@DESKTOP-QAIDGDP MINGW64 /e/programming/AI_ka_chilla_assignments (master)
$ git commit -m "added all folders"
[master (root-commit) bcf714e] added all folders
 10 files changed, 2010 insertions(+)
 create mode 100644 00_assignments.md
 create mode 100644 01_jupiter.ipynb
 create mode 100644 02_python_learning_complete.ipynb
 create mode 100644 03_markdown_practice.md
 create mode 100644 03_markdown_practice.pdf
 create mode 100644 screen_shots/bash.png
 create mode 100644 screen_shots/conda.png
 create mode 100644 screen_shots/git.png
 create mode 100644 screen_shots/jupiter.png
 create mode 100644 screen_shots/pic.jpg
(base)
UMS@DESKTOP-QAIDGDP MINGW64 /e/programming/AI_ka_chilla_assignments (master)
$ git remote add origin https://github.com/logicalzain/AI_ka_chilla_assignment.git
git branch -M main
git push -u origin main
Enumerating objects: 13, done.
Counting objects: 100% (13/13), done.
Delta compression using up to 8 threads
Compressing objects: 100% (13/13), done.
Writing objects: 100% (13/13), 325.44 KiB | 14.79 MiB/s, done.
Total 13 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), done.
To https://github.com/logicalzain/AI_ka_chilla_assignment.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
(base)
UMS@DESKTOP-QAIDGDP MINGW64 /e/programming/AI_ka_chilla_assignments (main)
$ 
```

Stage, commit, push in VSCode:





5. **Pandas, Numpy, Seaborn, Matplot & Plotly Practice with Custom Dataset**

Custom_Dataset
Data_Visualization

6. **EDA & Pandas-AI with API-keys with Custom Dataset**

Custom_Dataset
Pandas_AI
EDA_Pandas HTML_EDA

7. **Project- Currency Rate Comparison (2000-2026)**

Currency_Rate_Comparison

data sets:

Daily_Rate_Comparison
Yearly_Rate_Comparison

8. **Project- Streamlit Applications using AI**

API_application

9. **Ollama Chatbot in PC offline**

```
MINGW64:/c/Users/UMS
(base)
UMS@DESKTOP-QAIDGDP MINGW64 ~
$ ollama --version
ollama version is 0.18.0
(base)
UMS@DESKTOP-QAIDGDP MINGW64 ~
$ ollama list
NAME                    ID              SIZE      MODIFIED
llama2-uncensored:7b    44040b922233    3.8 GB    20 minutes ago

One important discovery that has helped shed light on this subject is the cosmic
microwave background radiation (CMBR). This is a form of electromagnetic
radiation that was emitted shortly after the Big Bang, providing us with
valuable information about the early universe. By analyzing the
temperature and polarization patterns in the CMBR, scientists have been
able to determine the properties of the universe at very high temperatures
and densities during its initial expansion phase.

Another important discovery is the existence of dark matter and energy.
These two components are believed to make up a significant portion of the
total mass-energy content of the universe, yet they interact with light
and other forms of electromagnetic radiation in ways that cannot be detected directly using

>>>
```

10. **Machine Learning practice**

Machine_Learning_Practice.

11. **AI Agent Project**

AI_Agent_Project

Final Project Documentation & Presentation

- Documentation
- Code